

EXPERIMENT 5

Spam Email Detection using Binary Text Classification

1. Problem Definition

The objective of this experiment is to develop a **spam email detection system** using multiple machine learning classification algorithms and compare their performance.

Spam detection is formulated as a **binary classification problem**, where:

- **1 → Spam**
- **0 → Ham (Not Spam)**

The goal is to accurately classify incoming emails and minimize:

- False Positives (important emails marked as spam)
- False Negatives (spam emails not detected)

2. Data Understanding

The dataset consists of email messages labeled into two categories:

- **Ham (Legitimate emails)**
- **Spam (Unwanted emails)**

Each record contains:

1. Message: raw email text
2. Category: label (spam/ham)

Dataset Statistics

- Total samples: **5572**
- Ham emails: **4825**
- Spam emails: **747**

Insight:

- Dataset is **imbalanced** (ham dominates)
- This directly affects evaluation → accuracy alone is misleading

3. Text Preprocessing

Raw text cannot be directly used in machine learning models. Therefore, **TF-IDF (Term Frequency – Inverse Document Frequency)** was used.

TF-IDF Concept

TF-IDF assigns importance to words using:

- **Term Frequency (TF)**
Frequency of a word in a document
- **Inverse Document Frequency (IDF)**
Measures rarity of a word across documents

Final representation:

Each email is converted into a **numerical feature vector**

This helps:

- Reduce noise (common words like “the”, “is” ignored)
- Highlight important keywords (e.g., “free”, “win”, “offer”)

4. Train–Test Split

The dataset was divided as:

- **80% Training Data**
- **20% Testing Data**

This ensures:

- Model learns on training data
- Performance is evaluated on unseen test data

5. Models Used

Five classification algorithms were implemented:

1. Naïve Bayes

- Based on Bayes Theorem
- Assumes feature independence

- Very effective for text classification

2. Logistic Regression

- Linear model for classification
- Uses sigmoid function
- Works well when data is linearly separable

3. Support Vector Machine (SVM)

- Finds optimal hyperplane
- Maximizes margin between classes
- Effective in high-dimensional spaces

4. Decision Tree

- Tree-based model
- Splits data using feature conditions
- Easy to interpret but prone to overfitting

5. Random Forest

- Ensemble of decision trees
- Uses bagging technique
- Reduces overfitting and improves generalization

6. Evaluation

To evaluate model performance, the following metrics were used:

Accuracy

Proportion of correctly classified samples

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Measures correctness of predicted spam

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

Measures ability to detect actual spam

F1-Score

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Balances precision and recall

ROC-AUC

Measures model's ability to distinguish between classes

Confusion Matrix

Breaks predictions into:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

Naïve Bayes

```
[[966 0]
 [ 25 124]]
```

- TN = 966 → correct ham classification

- $FP = 0 \rightarrow$ no false alarms
- $FN = 25 \rightarrow$ missed spam
- $TP = 124 \rightarrow$ correctly detected spam

Interpretation:

- Perfect precision (no ham misclassified)
- Some spam still missed

Random Forest

```
[[966 0]
 [ 22 127]]
```

Interpretation:

- Best recall among models
- Lowest missed spam
- Strong overall performance

7. Comparative Analysis

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Naïve Bayes	0.9776	1.0000	0.8322	0.9084	0.9853
Logistic Regression	0.9570	1.0000	0.6779	0.8080	0.9884
Support Vector Machine	0.9776	1.0000	0.8322	0.9084	0.9912
Decision Tree	0.9677	0.9124	0.8389	0.8741	0.9133
Random Forest	0.9803	1.0000	0.8523	0.9203	0.9921

Key Observations

1. High Precision Across Models

Most models achieve **precision = 1.0**

- No legitimate emails marked as spam
- Good for user experience

2. Recall is the Critical Metric

- Logistic Regression recall = **0.67 (poor)**
- Means many spam emails are missed

This makes it unsuitable despite good accuracy

3. Random Forest Performs Best

- Highest F1-score = **0.9203**
- Highest ROC-AUC = **0.9921**
- Best balance of precision and recall

4. SVM and Naïve Bayes Perform Similarly

- Same accuracy and F1-score
- SVM slightly better ROC performance

5. Decision Tree is Less Reliable

- Lower precision
- More classification errors